

Circles

Class - 8 Concise Mathematics Selina

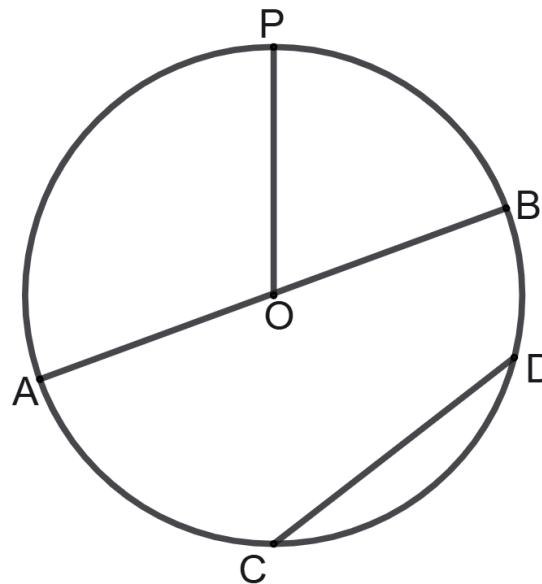
Three empty product cards are displayed in a row. Each card has a red discount tag in the top left corner: the first card shows '-45%', the second shows '-50%', and the third shows '-45%'. The cards are otherwise blank, with no product images or text visible.

Exercise 21

Question 1(i)

In the given figure, AB is the largest chord of the circle, $AB = 18$ cm and the length of the chord CD is half of length of chord AB, then :

Chapter 13
Factorisation
Chapter 14
Linear Equations in One Variable
Chapter 15
Linear Inequalities
Chapter 16
Understanding Shapes
Chapter 17
Special Types of Quadrilaterals
Chapter 18
Constructions
Chapter 19
Representing 3-D in 2-D
Chapter 20
Symmetry



1. $OP = 18$ cm and $CD = 18$ cm
2. $OP = 9$ cm and $CD = 18$ cm
3. $OP = 18$ cm and $CD = 9$ cm
4. $OP = 9$ cm and $CD = 9$ cm

Answer

As we know that

the largest chord of a circle is its diameter.

Diameter, $AB = 20$ cm

$$\text{Chord } CD = \text{half of length of chord } AB = \frac{1}{2} \times 18$$

= 9 cm.

OP is radius of the circle. So, radius is half of the diameter.

$$OP = \frac{1}{2} \times 18 = 9 \text{ cm}$$

$\therefore$ OP = 9 cm and CD = 9 cm

Hence, option 4 is the correct option.

Question 1(ii)

The largest chord of a circle is 20 cm, then its radius is :

1. 20 cm
2. 40 cm
3. 10 cm
4. none of these

Answer

The largest chord of a circle is its diameter.

$\therefore$ Diameter = 20 cm

As we know that radius is half of the diameter.

$$\text{Radius} = \frac{1}{2} \times 20 = 10 \text{ cm}$$

Hence, option 3 is the correct option.

Question 1(iii)

The relation between the area enclosed by a major segment of a circle and area enclosed by its semi-circle is :

1. major segment is smaller than the semi-circle.
2. major segment is bigger than the semi-circle.
3. major segment is equal to the semi-circle.

Answer

The central angle θ for the major segment is greater than 180° , leading to a larger sector area, and consequently, a larger segment area.

$\therefore$ Major segment is bigger than the semi-circle.

Hence, option 2 is the correct option.

Question 1(iv)

The radius of a circle is 5 cm then the length of its longest chord is :

1. less than 5 cm.
2. equal to 5 cm.
3. equal to 10 cm.
4. less than 10 cm

Answer

Given, radius of a circle = 5 cm.

The largest chord of a circle is its diameter.

As we know that radius is half of the diameter.

$$\Rightarrow \text{Diameter} = 2 \times \text{radius} = 2 \times 5 \text{ cm} = 10 \text{ cm}$$

$\therefore$ Length of its longest chord = 10 cm

Hence, option 3 is the correct option.

Question 1(v)

The diameter of a circle with centre at point O is 30 cm. A point P lies in the same plane as the circle and $OP = 20$ cm, then point P lies :

1. on the circumference of the circle
2. inside the circle.
3. outside the circle.
4. nothing can be said

Answer

Given, diameter of a circle = 30 cm.

As we know that radius is half of the diameter.

$$\Rightarrow \text{Radius} = \frac{1}{2} \times \text{diameter} = \frac{1}{2} \times 30$$

$$= 15 \text{ cm.}$$

The distance from the center O to point P is 20 cm. Since $OP = 20$ cm is greater than the radius of the circle (15 cm).

$\therefore$ Point P lies outside the circle.

Hence, option 3 is the correct option.

Question 1(vi)

The angle between a line AB and the diameter of a circle is 90° ; line AB is :

1. a tangent of the circle
2. not a tangent of the circle
3. nothing can be said

Answer

The angle between a line AB and the diameter of a circle is 90° ; line AB can be either a tangent of the circle or cannot be a tangent of the circle.

Thus, nothing can be confirmed.

Hence, option 3 is the correct option.

Question 1(vii)

A sector of a circle can be equal to its segment :

1. yes
2. no
3. none of these two

Answer

A segment of a circle is the area bounded by a chord of a circle and the area of the circle between the two endpoints of the chord.

A sector of a circle is a portion of a circle created by the arc of the circle and two radii.

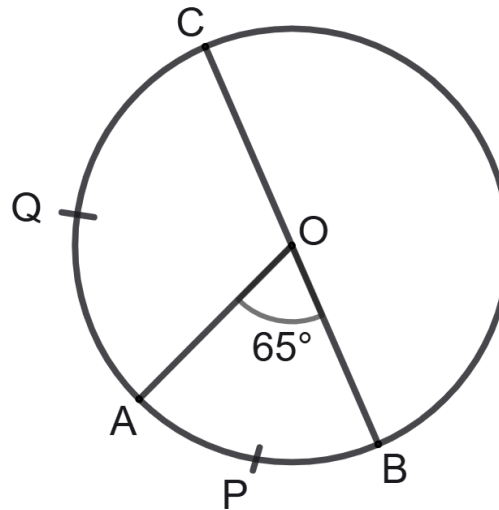
$\therefore$ A sector of a circle can be equal to a segment of the same circle when the sector is a semicircle.

Yes, a sector of a circle can be equal to its segment.

Hence, option 1 is the correct option.

Question 1(viii)

The given figure shows a circle with center O. If length of arc AQC is double of arc APB and $\angle AOB = 65^\circ$; then angle AOC is :



1. 65°

2. $2 \times 65^\circ$

3. $\frac{1}{2} \times 65^\circ$

4. none of these

Answer

Given, length of arc AQC is double of arc APB.

If arc AQC is twice the length of arc APB, then the angle subtended by arc AQC at the center ($\angle AOC$) will also be twice the angle subtended by arc APB ($\angle AOB$).

$$\therefore \angle AOC = 2 \times \angle AOB.$$

$$\Rightarrow \angle AOC = 2 \times 65^\circ$$

$$\Rightarrow \angle AOC = 130^\circ.$$

Hence, option 2 is the correct option.

Question 1(ix)

Statement 1: The angle between a radius and tangent of a circle is 90° .

Statement 2: At the point of contact the angle between radius and tangent is equal to one right angle.

Which of the following options is correct?

1. Both the statements are true.
2. Both the statements are false.
3. Statement 1 is true, and statement 2 is false.

4. Statement 1 is false, and statement 2 is true.

Answer

A tangent line touches the circle at one point. A radius connects the center to the circumference.

The angle between a radius and tangent at the point of contact is always 90° . This is a fundamental property of circles.

$\therefore$ Both the statements are true.

Hence, option 1 is the correct option.

Question 1(x)

Assertion (A) : The radius of a circle is 20 cm, then the length of its largest chord is also 20 cm.

Reason (R) : Longest chord of a circle is equal to its diameter.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

As we know that radius is half of the diameter.

Diameter = $2 \times \text{radius} = 2 \times 20 = 40$ cm.

We know that,

Longest chord of a circle is equal to its diameter.

So, reason (R) is true.

Longest chord = 40 cm

So, assertion (A) is false.

$\therefore$ A is false, but R is true.

Hence, option 4 is the correct option.

Question 1(xi)

Assertion (A) : When the radius of a circle is increased by 3 cm, its diameter will increase by 2×3 cm.

Reason (R) : Original radius is r cm, original diameter $2r$ cm

$\therefore$ Increase in length of diameter = $2(r + 3)$ cm - $2r$

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Original radius = r cm

Original diameter = $2r$ cm

$$\text{New radius} = (r + 3) \text{ cm}$$

$$\text{New diameter} = 2(r + 3) \text{ cm}$$

$$\text{Increase in length of diameter} = \text{New diameter} - \text{Original diameter}$$

$$= 2(r + 3) - 2r$$

$$= 2r + 2 \times 3 - 2r$$

$$= 2 \times 3$$

$$= 6 \text{ cm.}$$

$\therefore$ Both A and R are correct, and R is the correct explanation for A.

Hence, option 1 is the correct option.

Question 1(xii)

Assertion (A) : If a circle is of radius 8 cm and the length of its minor arc is also 8 cm, then the length of major arc = $8(2\pi - 1)$ cm.

Reason (R) : For every circle, sum of the length of its major arc and length of its minor arc is equal to the circumference of the circle.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Given, radius of a circle = 8 cm and the length of its minor arc = 8 cm

Circumference of the circle = $2\pi r = 2 \times \pi \times 8 = 16\pi$

We know that,

Length of minor arc + length of major arc = Circumference of the circle

So, reason (R) is true.

$$\Rightarrow 8 + \text{length of major arc} = 16\pi$$

$$\Rightarrow \text{length of major arc} = 16\pi - 8$$

$$\Rightarrow \text{length of major arc} = 8(2\pi - 1) \text{ cm}$$

So, assertion (A) is true.

$\therefore$ Both A and R are correct, and R is the correct explanation for A.

Hence, option 1 is the correct option.

Question 1(xiii)

Assertion (A) : A, B and C are three points on the circumference of a circle such that BC is diameter of the circle. If AC = 4 cm and AB = 3 cm, then diameter BC = 4 cm + 3 cm.

Reason (R) : Angle BAC = angle of semi-circle = 90° and so $BC^2 = AB^2 + AC^2$

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.

4. A is false, but R is true.

Answer

Given, AC = 4 cm and AB = 3 cm

Since BC is the diameter, angle BAC is a right angle.

Triangle ABC is a right triangle with hypotenuse BC.

Using Pythagoras theorem,

$$BC^2 = AB^2 + AC^2$$

So, reason (R) is true.

$$\Rightarrow BC^2 = 3^2 + 4^2$$

$$\Rightarrow BC^2 = 9 + 16$$

$$\Rightarrow BC^2 = 25$$

$$\Rightarrow BC = \sqrt{25}$$

$$\Rightarrow BC = 5 \text{ cm.}$$

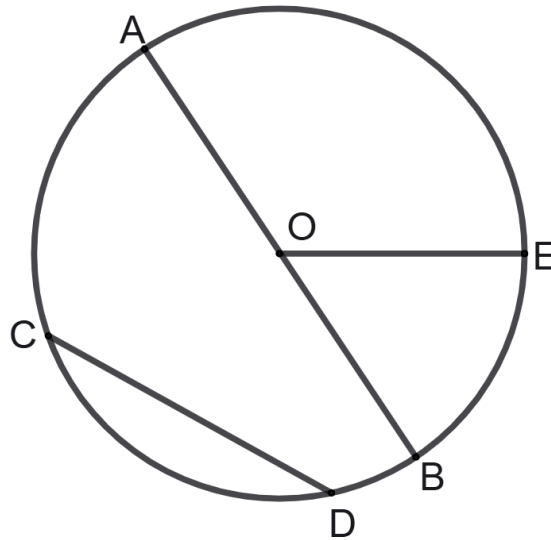
So, assertion (A) is false.

$\therefore$ A is false, but R is true.

Hence, option 4 is the correct option.

Question 2

The following figure shows the circle with centre O.



Use the figure to fill the blanks in each of the following :

- (i) $AB = \dots\dots\dots$
- (ii) Radius = $\dots\dots\dots$
- (iii) Chords = $\dots\dots\dots$ and $\dots\dots\dots$
- (iv) Diameter = $\dots\dots\dots$
- (v) $AB = 2 \times \dots\dots\dots$

Answer

- (i) $AB = \text{Diameter}$
- (ii) Radius = $OE = OA = OB$
- (iii) Chords = CD and AB

(iv) Diameter = **AB**

(v) $AB = 2 \times OE = OA = OB$

Question 3

M is a fixed point in a plane and a point P moves in the same plane such that $PM = 10$ cm. State :

- (i) the name of the figure formed.
- (ii) the length of the radius of the circle.
- (iii) the length of the diameter of the circle.

Can a chord of length 16 cm can be drawn in this circle? Give reason.

Answer

- (i) Given, M is a fixed point. P moves such that $PM = 10$ cm.

A circle is a set of points equidistant from a center or a fixed point.

Hence, the figure is circle with M as center.

- (ii) M is the center of the circle and PM is the radius of the circle.

Hence, the length of the radius of the circle = 10 cm.

- (iii) As we know that the diameter is twice the radius.

$$\text{Diameter} = 2 \times 10 \text{ cm} = 20 \text{ cm}$$

Hence, the length of the diameter of the circle = 20 cm.

The longest chord is the diameter.

The diameter is 20 cm.

A chord of length 16 cm can be drawn since $16 < 20$.

Hence, a chord of length 16 cm can be drawn in this circle.

Question 4

The radius of a circle is 6 cm. Find its diameter. If O is the centre of the circle; state, giving reasons, the position of points A, B and C; if;

(i) $OA = 4.8$ cm

(ii) $OB = 7.5$ cm

(iii) $OC = 6$ cm

Answer

The radius of a circle = 6 cm

As, the diameter is twice the radius.

Diameter = 2×6 cm = 12 cm.

(i) $OA = 4.8$ cm

As, $4.8 < 6$ (Radius)

Hence, A lies inside the circle.

(ii) $OB = 7.5$ cm

As, $7.5 > 6$ (Radius)

Hence, B lies outside the circle.

(iii) $OC = 6 \text{ cm}$

As, $6 = 6$ (Radius)

Hence, C lies on the circumference of the circle.

Question 5

Fill in the blanks :

- (i) An arc is the part of the
- (ii) Diameter of a circle bisects
- (iii) The part of the circumference greater than the semicircle is called
- (iv) Sector of a circle is its region bounded by
- (v) The segment of a circle is the region bounded by
- (vi) A tangent of a circle meets the circle at
- (vii) The number of tangents that can be drawn through a point on its circumference =
- (viii) The number of tangent that can be drawn through a point outside the circle is

Answer

- (i) An arc is a part of the **circumference of the circle**.
- (ii) Diameter of a circle bisects **circumference of the circle**.

- (iii) The part of the circumference greater than the semicircle is called **major-arc**.
- (iv) Sector of a circle is its region bounded by **two radii and an arc**.
- (v) The segment of a circle is the region bounded by **a chord and an arc**.
- (vi) A tangent of a circle meets the circle at **one point only**.
- (vii) The number of tangents that can be drawn through a point on its circumference = **only one**.
- (viii) The number of tangent that can be drawn through a point outside the circle is **two**.

Question 6

O is the center of a circle with diameter 30 cm. P is a point outside the circle and PA is tangent of the circle. Find :

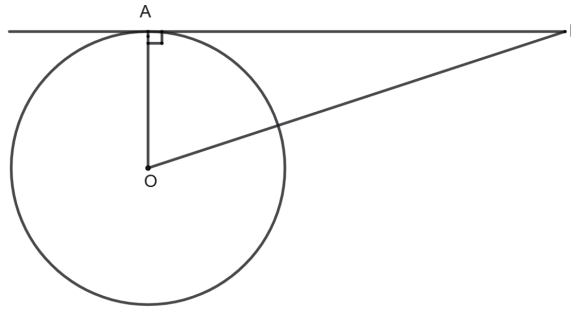
- (i) the length of tangent PA; if OP = 39 cm.
- (ii) the distance between O and P, if the length of the tangent PA is 20 cm.

Answer

Given, diameter = 30 cm

As we know that the diameter is twice the radius.

$$\Rightarrow \text{Radius} = \frac{\text{Diameter}}{2} = \frac{30}{2} = 15 \text{ cm.}$$



(i) It is given that $OP = 39$ cm.

In $\triangle OPA$,

$$\angle OAP = 90^\circ.$$

By pythagoras theorem,

$$\Rightarrow OA^2 + PA^2 = OP^2$$

$$\Rightarrow 15^2 + PA^2 = 39^2$$

$$\Rightarrow 225 + PA^2 = 1521$$

$$\Rightarrow PA^2 = 1521 - 225$$

$$\Rightarrow PA^2 = 1296$$

$$\Rightarrow PA = \sqrt{1296}$$

$$\Rightarrow PA = 36 \text{ cm.}$$

Hence, the length of tangent $PA = 36$ cm.

(ii) Given,

$$PA = 20 \text{ cm}$$

$$\Rightarrow OA^2 + PA^2 = OP^2$$

$$\Rightarrow 15^2 + 20^2 = OP^2$$

$$\Rightarrow 225 + 400 = OP^2$$

$$\Rightarrow OP^2 = 625$$

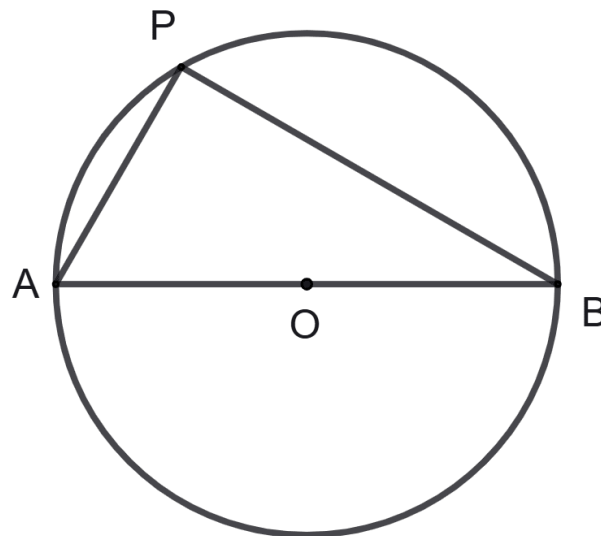
$$\Rightarrow OP = \sqrt{625}$$

$$\Rightarrow OP = 25 \text{ cm.}$$

Hence, the distance between O and P = 25 cm.

Question 7

The following figure shows a circle with center O and a diameter AB.



- (i) Name the angle APB.
- (ii) State the measure of angle APB.
- (iii) If $AP = 12$ cm and $OA = 10$ cm; find the lengths of AB and BP.
- (iv) If $4AP = 3PB = 12$ cm; find the radius of the circle.

Answer

- (i) AB is diameter of the circle.

Hence, angle APB is the angle of semi-circle.

- (ii) As, the angle subtended by a diameter at any point on the circumference of a circle is a right angle (90°).

Hence, angle APB = 90° .

- (iii) Given,

$$AP = 12 \text{ cm and } OA = 10 \text{ cm}$$

$$\Rightarrow OA = OB \text{ (Radii of the circle)}$$

$$\Rightarrow OB = 10 \text{ cm}$$

From figure,

$$\Rightarrow AB = OA + OB = 10 \text{ cm} + 10 \text{ cm} = 20 \text{ cm}$$

Since, angle APB = 90° .

Using pythagoras theorem,

$$\Rightarrow AP^2 + PB^2 = AB^2$$

$$\Rightarrow 12^2 + PB^2 = 20^2$$

$$\Rightarrow 144 + PB^2 = 400$$

$$\Rightarrow PB^2 = 400 - 144$$

$$\Rightarrow PB^2 = 256$$

$$\Rightarrow PB = \sqrt{256}$$

$$\Rightarrow PB = 16 \text{ cm.}$$

Hence, AB = 20 cm and PB = 16 cm.

(iv) Given,

$$\Rightarrow 4AP = 3PB = 12 \text{ cm}$$

$$\Rightarrow AP = \frac{12}{4} = 3 \text{ cm}$$

$$\Rightarrow PB = \frac{12}{3} = 4 \text{ cm}$$

Since, angle APB = 90° .

Using pythagoras theorem,

$$\Rightarrow AP^2 + PB^2 = AB^2$$

$$\Rightarrow 3^2 + 4^2 = AB^2$$

$$\Rightarrow 9 + 16 = AB^2$$

$$\Rightarrow 25 = AB^2$$

$$\Rightarrow AB = \sqrt{25}$$

$$\Rightarrow AB = 5 \text{ cm}$$

Now, $OA = OB$ (Radii of the circle)

From figure,

$$\Rightarrow AB = OA + OB$$

$$\Rightarrow AB = 2 \times OA$$

$$\Rightarrow 5 = 2 \times OA$$

$$\Rightarrow OA = \frac{5}{2} = 2.5 \text{ cm.}$$

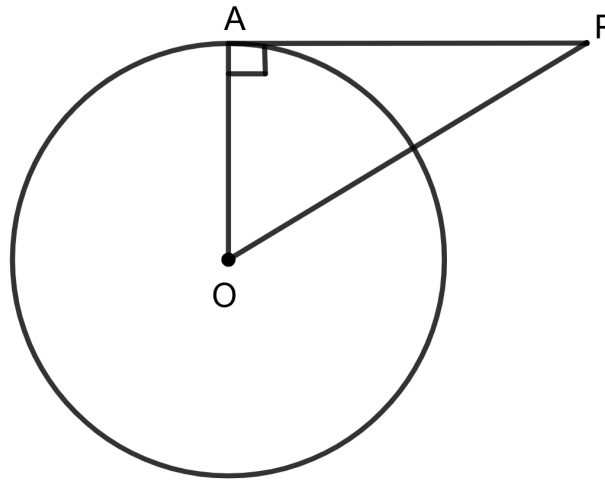
Hence, the radius of the circle = 2.5 cm.

Question 8

Find the length of the tangent to a circle with radius 5 cm, from a point at a distance of 13 cm from its center.

Answer

Consider a circle with center O. Let AP is the tangent of the circle.



We know that,

Tangent and radius to the circle at point of contact are perpendicular to each other.

In the $\triangle OPA$,

$$\angle OAP = 90^\circ.$$

By pythagoras theorem,

$$\Rightarrow OA^2 + PA^2 = OP^2$$

$$\Rightarrow 5^2 + PA^2 = 13^2$$

$$\Rightarrow 25 + PA^2 = 169$$

$$\Rightarrow PA^2 = 169 - 25$$

$$\Rightarrow PA^2 = 144$$

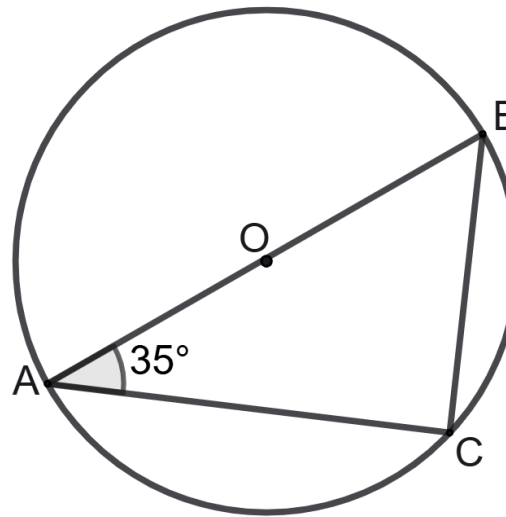
$$\Rightarrow PA = \sqrt{144}$$

$$\Rightarrow PA = 12 \text{ cm}$$

Hence, the length of the tangent = 12 cm.

Question 9

The following figure shows a circle with diameter AB and center at point O.



If angle CAB = 35°, find the measure of angle ABC.

Answer

We know that,

Angle in a semi-circle is a right angle.

So, angle ACB = 90°.

In $\triangle ACB$,

Using angle sum property,

$$\Rightarrow \angle ABC + \angle ACB + \angle CAB = 180^\circ$$

$$\Rightarrow \angle ABC + 90^\circ + 35^\circ = 180^\circ$$

$$\Rightarrow \angle ABC + 125^\circ = 180^\circ$$

$$\Rightarrow \angle ABC = 180^\circ - 125^\circ$$

$$\Rightarrow \angle ABC = 55^\circ.$$

Hence, angle ABC = 55°.

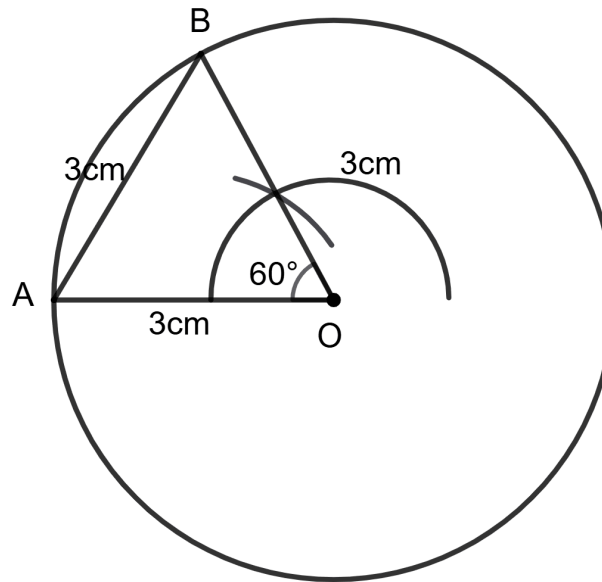
Question 10

Draw a circle with radius 3 cm and centre at point O. Draw two radii OA and OB of the circle drawn such that $\angle AOB = 60^\circ$. Join A and B and measure the length of AB.

Answer

Steps of construction :

1. Mark a point O.
2. With O as center and radius equal to 3 cm draw a circle.
3. Draw OA as the radius of the circle.
4. Draw another radius OB at an angle of 60° with OA.
5. Join A and B. Measure AB.



Hence, the length of $AB = 3$ cm.

Question 11

Draw a circle with radius 4 cm. Draw a chord AB of length 6 cm. Shade its minor segment by horizontal lines and major segment by vertical lines.

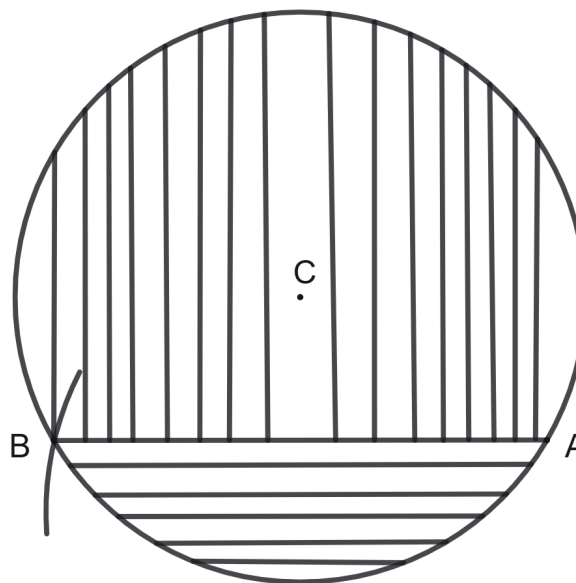
Answer

Steps of construction:

1. Draw a circle of radius 4 cm and mark the centre as C .
2. Mark a point A on the circle.
3. Draw an arc with A as centre and radius 6 cm to cut the circle at B .
4. Join the line AB .

5. Shade the segment which contains the centre 'C' by vertical lines.

6. Shade the segment which does not contain the centre 'C' by horizontal lines.



Question 12

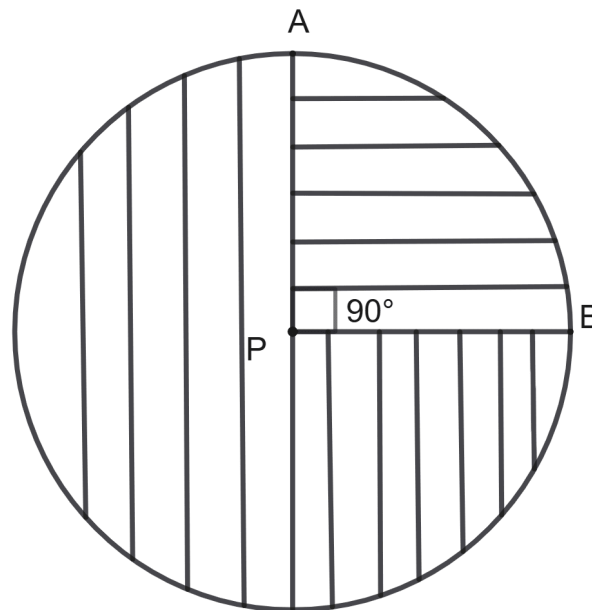
Draw a circle with centre at point P. Draw its radii PA and PB such that angle $APB = 90^\circ$. Shade the minor sector of the circle by horizontal lines and its major sector by vertical lines.

Answer

Steps of construction:

1. Draw a circle of any radius and mark the centre as P.
2. Draw a straight line PB.

3. Draw an angle APB such that angle APB = 90° , where A is a point on the circumference of the circle.
4. Shade the major segment by vertical lines.
5. Shade the minor segment by horizontal lines.

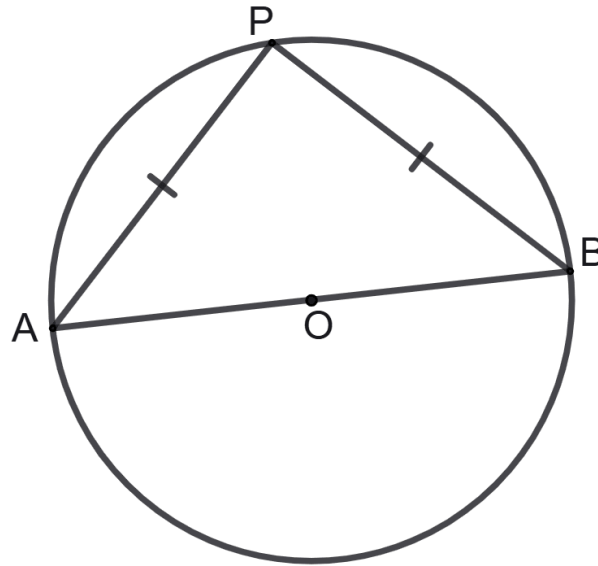


Question 13

In a circle with centre O and diameter AB; angle APB is the angle of semicircle.
If $PA = PB$; find the measure of each angle of triangle APB.

Answer

Given, O is center of circle. AB is the diameter of the circle.



$\Rightarrow PA = PB$ (Given)

$\Rightarrow \angle PAB = \angle PBA = x$ (let) [Angles opposite to equal sides of a triangle are always equal]

We know that,

Angle in a semi-circle is a right angle.

$\Rightarrow \angle APB = 90^\circ$

In $\triangle APB$, using angle sum property

$\Rightarrow \angle PAB + \angle PBA + \angle APB = 180^\circ$

$\Rightarrow x + x + 90^\circ = 180^\circ$

$\Rightarrow 2x = 180^\circ - 90^\circ$

$\Rightarrow 2x = 90^\circ$

$$\Rightarrow x = \frac{90^\circ}{2}$$

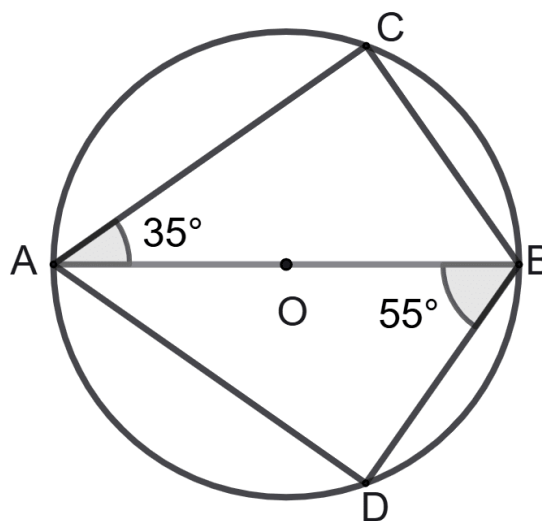
$$\Rightarrow x = 45^\circ$$

$$\Rightarrow \angle PAB = \angle PBA = 45^\circ$$

Hence, all angles of the triangles are 45° , 45° and 90° .

Question 14

In the following figure, O is the centre of the circle and AB is a diameter.



C and D are the points on the circumference of the circle such that $\angle CAB = 35^\circ$ and $\angle ABD = 55^\circ$. Find the measures of angles CAD and CBD.

Answer

We know that,

Angle in a semi-circle is a right angle.

$$\Rightarrow \angle ACB = \angle ADB = 90^\circ$$

In $\triangle ACB$, using angle sum property

$$\Rightarrow \angle CAB + \angle CBA + \angle ACB = 180^\circ$$

$$\Rightarrow 35^\circ + \angle CBA + 90^\circ = 180^\circ$$

$$\Rightarrow 125^\circ + \angle CBA = 180^\circ$$

$$\Rightarrow \angle CBA = 180^\circ - 125^\circ$$

$$\Rightarrow \angle CBA = 55^\circ$$

From figure,

$$\Rightarrow \angle CBD = \angle CBA + \angle ABD = 55^\circ + 55^\circ = 110^\circ.$$

Similarly, in $\triangle ADB$, using angle sum property

$$\Rightarrow \angle DAB + \angle DBA + \angle ADB = 180^\circ$$

$$\Rightarrow \angle DAB + 55^\circ + 90^\circ = 180^\circ$$

$$\Rightarrow \angle DAB + 145^\circ = 180^\circ$$

$$\Rightarrow \angle DAB = 180^\circ - 145^\circ$$

$$\Rightarrow \angle DAB = 35^\circ$$

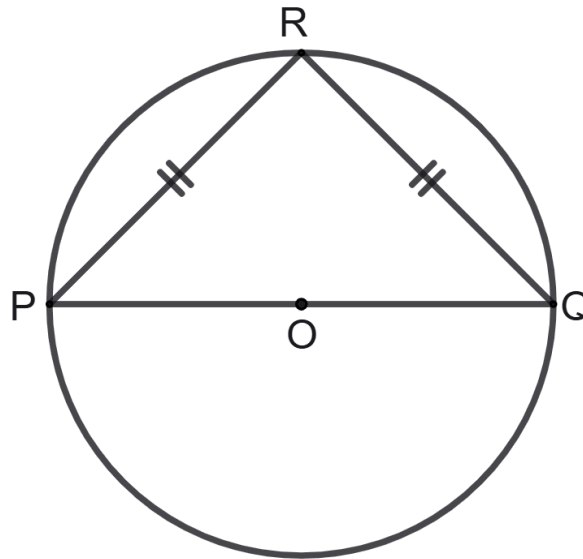
From figure,

$$\Rightarrow \angle CAD = \angle CAB + \angle DAB = 35^\circ + 35^\circ = 70^\circ.$$

Hence, angles $CAD = 70^\circ$ and angle $CBD = 110^\circ$.

Question 15

The following figure shows a circle with centre O and diameter PQ. Point R lies on the circumference of the circle such that $PR = QR$ and $PR = 4$ cm.



Calculate the radius of the given circle and state its value correct to two decimal places.

Answer

Given, PQ is diameter of the circle.

$PR = QR$ and $PR = 4$ cm

$\Rightarrow QR = 4$ cm

As we know that,

Angle in a semi-circle is a right angle.

$$\Rightarrow \angle PRQ = 90^\circ$$

Triangle PQR is a right triangle with hypotenuse PQ. Using Pythagoras theorem,

$$\Rightarrow PQ^2 = PR^2 + RQ^2$$

$$\Rightarrow PQ^2 = 4^2 + 4^2$$

$$\Rightarrow PQ^2 = 16 + 16$$

$$\Rightarrow PQ^2 = 32$$

$$\Rightarrow PQ = \sqrt{32} = 5.66 \text{ cm}$$

$$\text{Radius (OP)} = \frac{1}{2} \times PQ = \frac{1}{2} \times 5.66$$

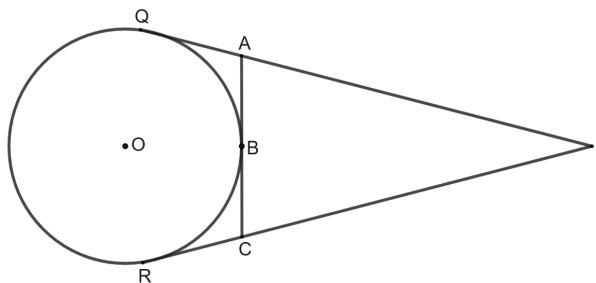
$$= 2.83 \text{ cm.}$$

Hence, the radius of the given circle = 2.83 cm.

Question 16

In the following figure, PQ and PR are the tangent to the circle with centre O.

Line segment AC touch the circle at point B.



(i) State the relation between tangent PQ and PR.

Also show that:

(ii) $PQ = PA + AB$

(iii) $PR = PC + CB$

(iv) $PQ + PR = \text{Perimeter of } \Delta PAC.$

Answer

(i) PQ and PR are the tangent to the circle with centre O.

As we know that,

Through any point, outside a circle at most two tangent can be drawn and both of these tangent are always equal to each other.

Here, point P lies outside the circle and the two tangent PQ and PR are drawn to the circle.

Hence, $PQ = PR$.

(ii) When point A lies outside the circle and the two tangent AQ and AB are drawn to the circle. Then,

$AQ = AB \dots\dots\dots(1)$

From figure,

$\Rightarrow PQ = PA + AQ$

From equation (1),

$\Rightarrow PQ = PA + AB$

Hence, proved that $PQ = PA + AB$.

(iii) When point C lies outside the circle and the two tangent CR and CB are drawn to the circle. Then

$$CR = CB \dots\dots\dots(2)$$

From figure,

$$\Rightarrow PR = PC + CR$$

From equation (2),

$$\Rightarrow PR = PC + CB$$

Hence, proved that $PR = PC + CB$.

(iv) Perimeter of triangle PAC = PA + PC + AC

$$= PA + PC + (AB + CB)$$

$$= PA + PC + (AQ + CR)$$

$$= (PA + AQ) + (PC + CR)$$

$$= PQ + PR$$

Hence, proved that $PQ + PR = \text{Perimeter of } \Delta PAC$.

Question 17

From an external point P, the tangent PA is drawn to the circle with centre O.

(i) If OP = 20 cm and tangent PA = 16 cm find the diameter of the circle.

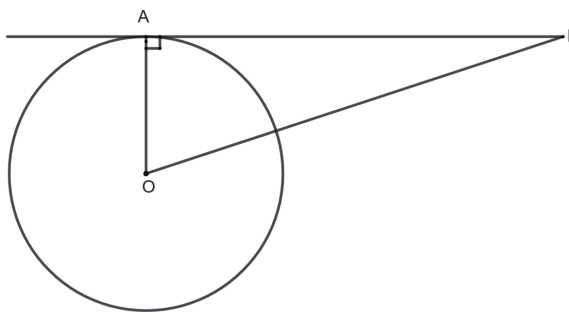
(ii) If diameter of the circle is 20 cm and tangent $PA = 24$ cm, find the length of OP .

Answer

O is center of the circle. PA is the tangent of the circle.

We know that,

Tangent and radius through the center at point of contact are perpendicular to each other.



(i) Given, $OP = 20$ cm and $PA = 16$ cm

In the $\triangle OPA$ as we know that $\angle OAP = 90^\circ$.

As we know Pythagoras theorem can be used in a right angled triangle,

$$\Rightarrow OA^2 + PA^2 = OP^2$$

$$\Rightarrow OA^2 + 16^2 = 20^2$$

$$\Rightarrow OA^2 + 256 = 400$$

$$\Rightarrow OA^2 = 400 - 256$$

$$\Rightarrow OA^2 = 144$$

$$\Rightarrow OA = \sqrt{144}$$

$$\Rightarrow OA = 12 \text{ cm}$$

By formula,

$$\text{Diameter} = 2 \times \text{radius} = 2 \times 12 = 24 \text{ cm.}$$

Hence, the diameter of the circle = 24 cm.

(ii) Given, diameter = 20 cm and PA = 24 cm

$$\text{Radius (OA)} = \frac{1}{2} \times 20 = 10 \text{ cm}$$

In the $\triangle OPA$ as we know that $\angle OAP = 90^\circ$.

As we know Pythagoras theorem can be used in a right angled triangle,

$$\Rightarrow OA^2 + PA^2 = OP^2$$

$$\Rightarrow 10^2 + 24^2 = OP^2$$

$$\Rightarrow 100 + 576 = OP^2$$

$$\Rightarrow OP^2 = 676$$

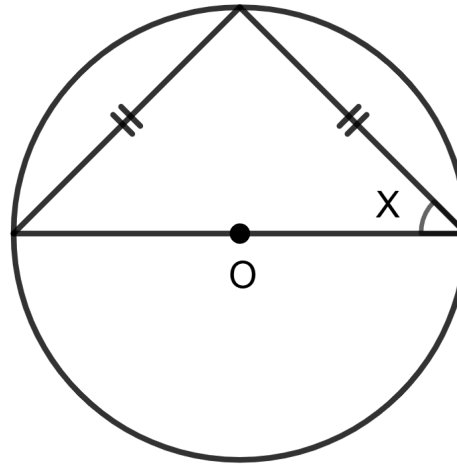
$$\Rightarrow OP = \sqrt{676}$$

$$\Rightarrow OP = 26 \text{ cm.}$$

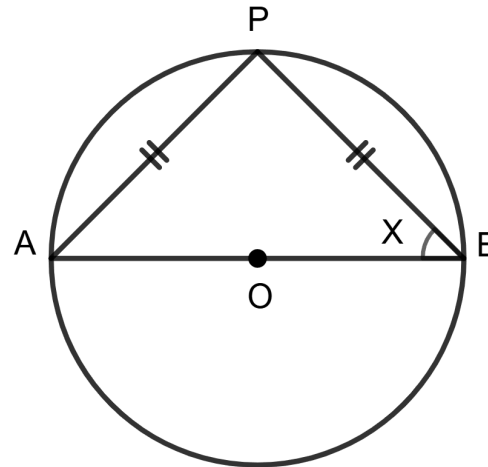
Hence, the length of OP = 26 cm.

Question 18(i)

In figure given below, O is the centre of the circle, use the given information to find the value of x:



Answer



PA = PB (Given)

$\Rightarrow \angle PAB = \angle PBA = x$ (let) [Angles opposite to equal sides of a triangle are always equal]

As we know that the angle subtended by a diameter at any point on the circumference of a circle is a right angle (90°).

$$\Rightarrow \angle APB = 90^\circ$$

In $\triangle APB$, using angle sum property

$$\Rightarrow \angle PAB + \angle PBA + \angle APB = 180^\circ$$

$$\Rightarrow x + x + 90^\circ = 180^\circ$$

$$\Rightarrow 2x = 180^\circ - 90^\circ$$

$$\Rightarrow 2x = 90^\circ$$

$$\Rightarrow x = \frac{90^\circ}{2}$$

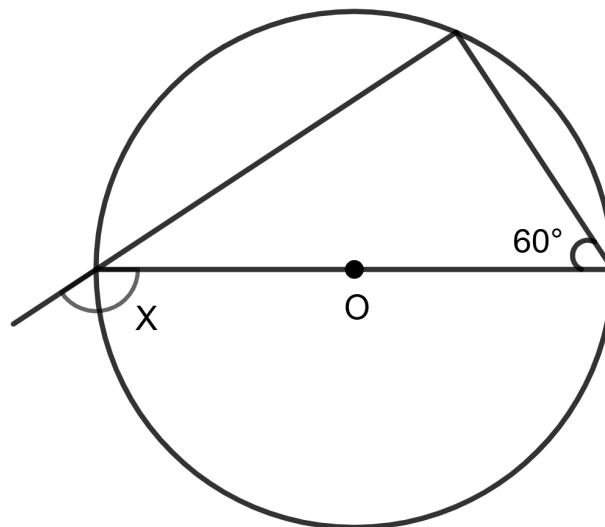
$$\Rightarrow x = 45^\circ$$

$$\Rightarrow \angle PAB = \angle PBA = 45^\circ$$

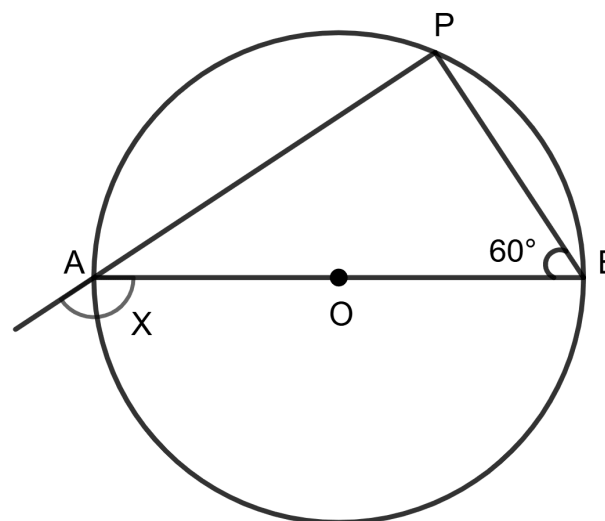
Hence, the value of $x = 45^\circ$.

Question 18(ii)

In figure given below, O is the centre of the circle, use the given information to find the value of x :



Answer



As we know that the angle subtended by a diameter at any point on the circumference of a circle is a right angle (90°).

$$\Rightarrow \angle APB = 90^\circ$$

According to exterior angle theorem,

Exterior angle of a triangle is equal to the sum of the two opposite interior angles.

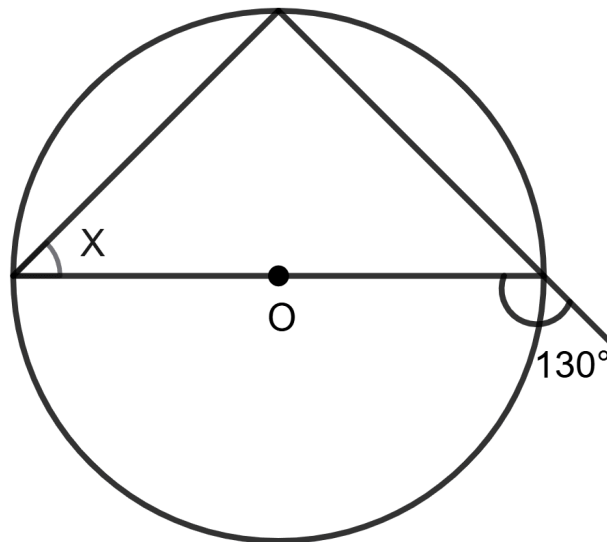
$$\Rightarrow \text{ex. } \angle A = \angle APB + \angle ABP$$

$$\Rightarrow x = 90^\circ + 60^\circ = 150^\circ$$

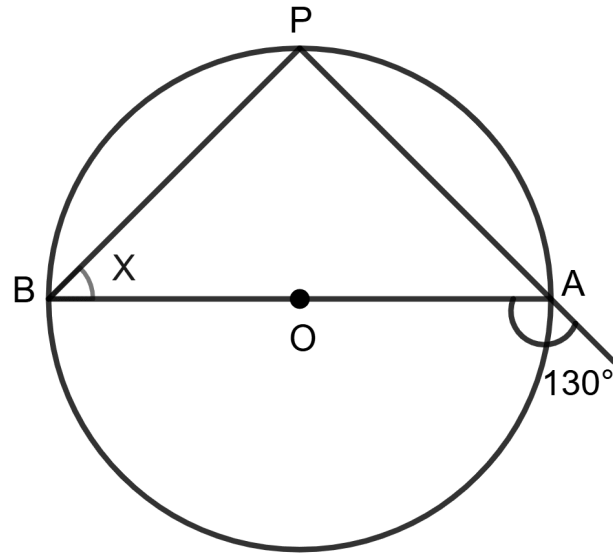
Hence, the value of $x = 150^\circ$.

Question 18(iii)

In figure given below, O is the centre of the circle, use the given information to find the value of x :



Answer



As we know that the angle subtended by a diameter at any point on the circumference of a circle is a right angle (90°).

$$\Rightarrow \angle APB = 90^\circ$$

According to exterior angle theorem,

Exterior angle of a triangle is equal to the sum of the two opposite interior angles.

$$\Rightarrow \text{ex. } \angle A = \angle APB + \angle ABP$$

$$\Rightarrow 130^\circ = 90^\circ + x$$

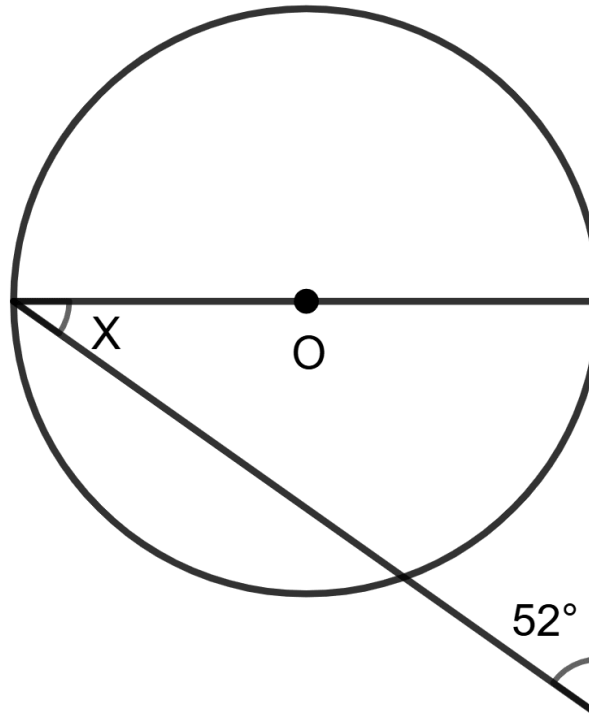
$$\Rightarrow x = 130^\circ - 90^\circ$$

$$\Rightarrow x = 40^\circ.$$

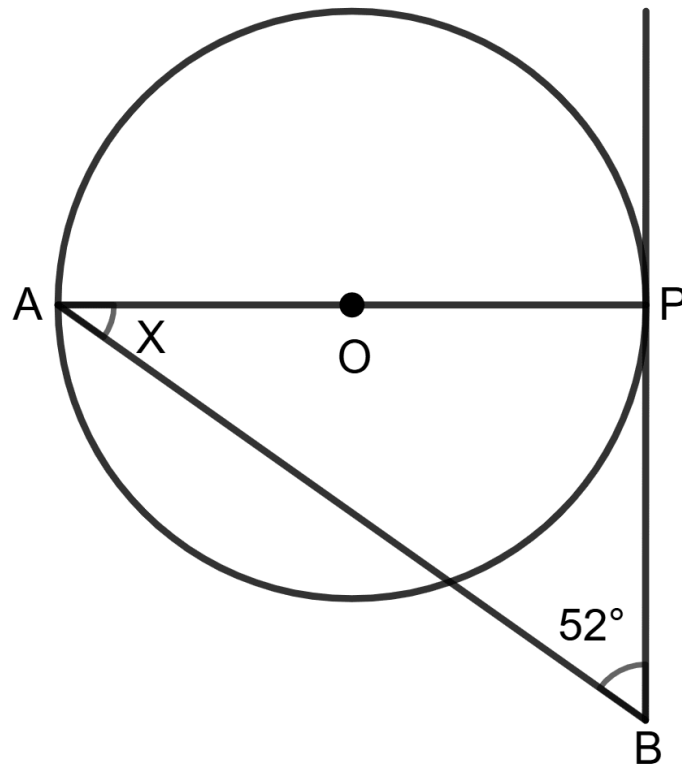
Hence, the value of $x = 40^\circ$.

Question 18(iv)

In figure given below, O is the centre of the circle, use the given information to find the value of x:



Answer



At the point of contact, the angle between tangent and the radius is always 90° .

$$\Rightarrow \angle APB = 90^\circ$$

In $\triangle APB$, using angle sum property

$$\Rightarrow \angle PAB + \angle PBA + \angle APB = 180^\circ$$

$$\Rightarrow x + 52^\circ + 90^\circ = 180^\circ$$

$$\Rightarrow x + 142^\circ = 180^\circ$$

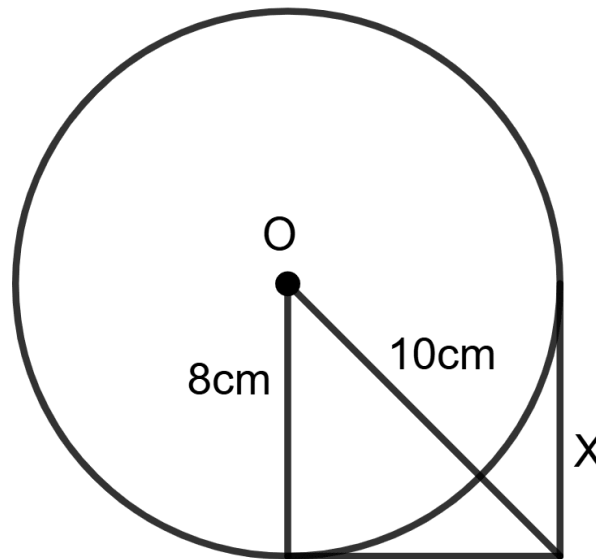
$$\Rightarrow x = 180^\circ - 142^\circ$$

$$\Rightarrow x = 38^\circ.$$

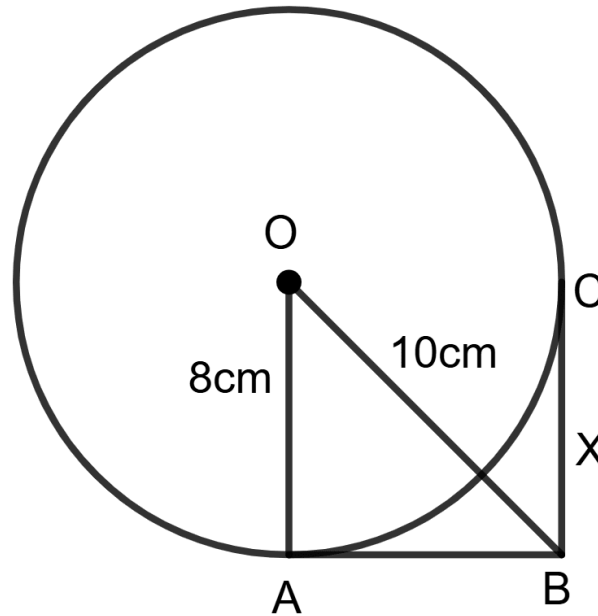
Hence, the value of $x = 38^\circ$.

Question 18(v)

In figure given below, O is the centre of the circle, use the given information to find the value of x :



Answer



At the point of contact, the angle between tangent and the radius is always 90° .

$$\Rightarrow \angle OAB = 90^\circ$$

Triangle OAB is a right triangle with hypotenuse OB. Using Pythagoras theorem,

$$\Rightarrow OB^2 = OA^2 + AB^2$$

$$\Rightarrow 10^2 = 8^2 + AB^2$$

$$\Rightarrow 100 = 64 + AB^2$$

$$\Rightarrow AB^2 = 100 - 64$$

$$\Rightarrow AB^2 = 36$$

$$\Rightarrow AB = \sqrt{36}$$

$$\Rightarrow AB = 6 \text{ cm}$$

We know that,

Tangents from an exterior point to a circle are equal in length.

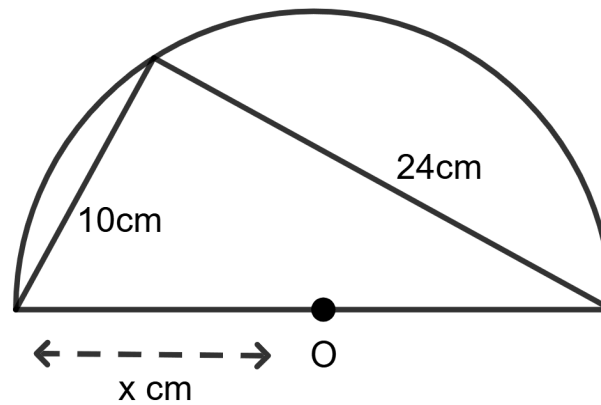
$$\Rightarrow BC = BA$$

$$\Rightarrow x = 6 \text{ cm.}$$

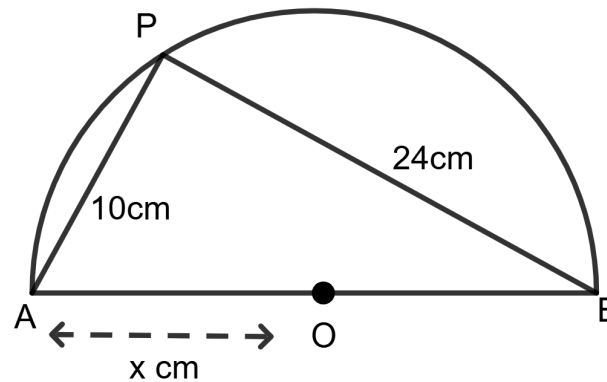
Hence, the value of $x = 6 \text{ cm}$.

Question 18(vi)

In figure given below, O is the centre of the circle, use the given information to find the value of x :



Answer



Angle in semi-circle is a right angle.

$$\Rightarrow \angle APB = 90^\circ$$

Triangle APB is a right triangle with hypotenuse AB. Using Pythagoras theorem,

$$\Rightarrow AB^2 = PA^2 + PB^2$$

$$\Rightarrow AB^2 = 10^2 + 24^2$$

$$\Rightarrow AB^2 = 100 + 576$$

$$\Rightarrow AB^2 = 676$$

$$\Rightarrow AB = \sqrt{676}$$

$$\Rightarrow AB = 26 \text{ cm}$$

Radius is half of the diameter.

$$\Rightarrow x = \frac{AB}{2}$$

$$\Rightarrow x = \frac{1}{2} \times 26 = 13 \text{ cm.}$$

Hence, the value of $x = 13\text{ cm}$.

[← Prev](#)
[Symmetry](#)

[Area of a Trapezium and a Polygon](#) [Next →](#)

Discover more

Colleges & Universities



TV & Video Equipment



Computers



ICSE/ISC TEXTBOOK SOLUTIONS

[Class 6 ICSE Textbook Solutions](#)

[Class 7 ICSE Textbook Solutions](#)

[Class 8 ICSE Textbook Solutions](#)

[Class 9 ICSE Textbook Solutions](#)

[Class 10 ICSE Textbook Solutions](#)

[Class 11 ISC Textbook Solutions](#)

[Class 12 ISC Textbook Solutions](#)

CBSE TEXTBOOK SOLUTIONS

[Class 8 CBSE Textbook Solutions](#)

[Class 9 CBSE Textbook Solutions](#)

[Class 10 CBSE Textbook Solutions](#)

[Class 11 CBSE Textbook Solutions](#)

[Class 12 CBSE Textbook Solutions](#)

ICSE/ISC/CBSE SOLVED QUESTION PAPERS

[ICSE Previous Year Question Papers Solved Class 10](#)

COMPANY

[Pricing](#)

[About Us](#)

[Contact Us](#)

[Privacy Policy](#)

[Terms of Service](#)

Selina Solutions for Class 10, 9, 8, 7, 6

Morning Star Solutions for Class 10, 9

ICSE/ISC PRACTICE TESTS

Class 8 ICSE Practice Tests

Class 9 ICSE Practice Tests

Class 10 ICSE Practice Tests

ICSE Previous Year Question Papers Solved Class 12

CBSE Previous Year Question Papers Solved Class 10

CBSE Previous Year Question Papers Solved Class 12

ICSE Competency Focussed Practice Questions Solved Class 10

CBSE PRACTICE TESTS

Class 10 CBSE Practice Tests

Class 12 CBSE Practice Tests



Copyright © KnowledgeBoat 2026